

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	6 September 2026
Team ID	Individual
Project Name	An employee requests installation of licensed software through the Service Catalog
Maximum Marks	4 Marks

Technical Architecture:

The proposed solution uses ServiceNow as the application platform. Employees submit a licensed software installation request through the Service Catalog. The request is stored as REQ/RITM, routed through Flow Designer for approval, and after approval a Catalog Task (SCTASK) is created for the IT/Software Team to complete the installation.

Architecture Flow

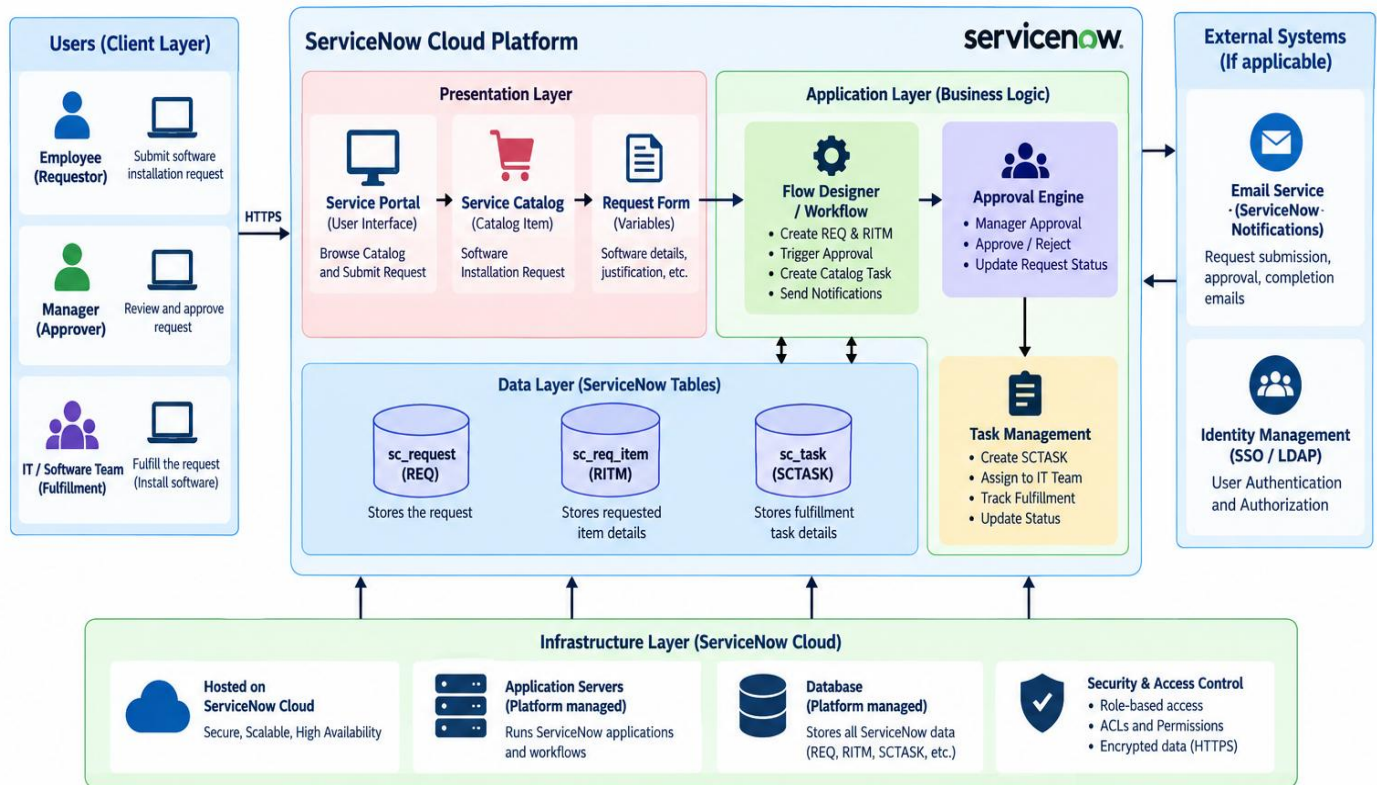


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interface through which the employee submits and tracks the software installation request.	ServiceNow Service Portal / Service Catalog
2	Application Logic-1	Captures software request details and creates the request records.	Service Catalog Item, Catalog Variables
3	Application Logic-2	Controls approval routing and process automation.	ServiceNow Flow Designer
4	Application Logic-3	Creates and manages the fulfillment task after approval.	Flow Designer, Catalog Task (SCTASK)
5	Database	Stores request, requested item, approval, and task information.	ServiceNow tables: sc_request, sc_req_item, sysapproval_approver, sc_task
6	Cloud Database	Platform-managed data storage for the ServiceNow application.	ServiceNow platform data store
7	File Storage	No separate file-storage component is required for the core project workflow.	Not applicable
8	External API-1	No external API is required for software request processing.	Not applicable
9	External API-2	No external API is required for software request processing.	Not applicable
10	Machine Learning Model	No machine-learning model is required for the proposed workflow.	Not applicable
11	Infrastructure (Server / Cloud)	Application and workflow are hosted and executed on the ServiceNow instance.	ServiceNow Personal Developer Instance / Enterprise Instance

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is required. The solution is configured within the ServiceNow platform.	ServiceNow platform configuration
2	Security	Access to catalog requests and approvals is controlled using ServiceNow roles and permissions. The workflow supports authorized approval before fulfillment.	ServiceNow roles, ACLs, authentication and authorization
3	Scalable Architecture	The solution is platform-based and can support additional software catalog items, users, approval rules, and fulfillment tasks as the request volume grows.	ServiceNow platform and Flow Designer
4	Availability	The application runs on the ServiceNow platform, allowing employees and IT teams to access the service through the configured instance.	ServiceNow instance
5	Performance	The workflow minimizes manual coordination by automating request routing, approval, and task creation. Catalog variables capture required information at submission.	Service Catalog and Flow Designer